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Predicting Repurchase Behavior and Optimizing Marketing for E-commerce Users with Genetic Algorithms and Deep Learning

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Abstract

The quick expansions of e-commerce have increased competition, made client retention and repurchased predictions vital for sustainable corporate success. Traditional Deep Learning (DL) algorithms frequently fall short of collecting complex behavioral patterns and optimizing marketing campaigns. Research provides a unique framework that mixes DL and Genetic Algorithms (GA) to predict purchasing patterns and improve precision marketing for e-commerce platforms. The data of predicting e-commerce user repurchase behavior is obtained. The data features are preprocessed through handling missing values by imputation, removing duplicates, and utilizing min-max normalization for determining numerical features. A **Dynamic Genetic Algorithm-mutated Enhanced Vanilla Memory Network (DGA-EVMN)** is employed to enable the accurate predictions of repurchase possibilities. The EVMN model provided the temporal dependencies and nonlinear relationships in customer repurchase behavior. To further refine business decision-making, a DGA is applied to optimize personalized marketing interventions such as targeted discounts, product recommendations, and promotional campaigns. Experimental evaluation on real-world e-commerce datasets demonstrates that the proposed DGA-EVMN framework significantly outperforms traditional baseline models, achieving improvements in prediction F1-score (0.960), AUC-ROC (0.971), recall (0.962), precision (0.958) and accuracy (0.984) by the Python implementation. Moreover, the DGA-driven optimization results in more

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efficient resource allocation, marketing waste reduction, and higher conversion rates. The findings highlight the potential of combining predictive analytics with evolutionary optimization to enable intelligent, data-driven marketing strategies, helping businesses to maximize customer lifetime value through adaptive and cost-effective practices.

Keywords: E-Commerce, Repurchase Behavior, Customer Retention, Marketing Optimization, Dynamic Genetic Algorithm-Mutated Enhanced Vanilla Memory Network (DGA-EVMN)

1. Introduction

Customers tend to save time and money while purchasing through the internet. Online purchases have evolved as an important consumption platform. Online buying has changed dramatically and it is gradually displacing traditional offline purchase methods, especially with the epidemic's effects on traditional businesses [1]. E-commerce contributes greatly to global commerce by simplifying corporate development into international markets, minimizing Business-To-Business (B2B) transactions, building faster worldwide networks, and experiences compared to conventional transactions [2-3]. Customer acquisition is critical, while retention is a more advantageous technique for long-term profitability. Comparative investigations are crucial in marketing considering men and women have different online shopping habits [4]. Repurchase intention is a significant indicator for business performance and has gained a lot of popularity in e-commerce investigations [5]. Establishing and maintaining satisfied customers through the preceding variables is a vital approach for increasing customer satisfaction [6]. The behavioral variables are expanded to the investigations on customers' repeated purchase intentions in farming-related e-commerce by the live transmission intentions while providing techniques for sellers to sell products effectively [7].

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A prominent establishment of online purchasing (E-commerce) modified the customer behavior by providing new options for businesses and sharpening competition [8]. Many models failed to completely reflect the effect of complicated temporal patterns, nonlinear behavioral dynamics [9]. Customers increasingly use e-commerce platforms to save time and money while purchasing. However, it is important to note that not all products are purchased online. Online shopping is primarily suited for ready-made or easily made-to-order items, whereas tailored or highly customized products such as those from tailors, carpenters, blacksmiths, watchmakers, or upholsterers typically require multiple human interactions and iterations and are less likely to be purchased online [10].

This research presents an integrated DL and evolutionary optimization framework that effectively combines established techniques to address e-commerce repurchase prediction and marketing optimization. By unifying predictive modeling with adaptive marketing strategies, the framework provides a practical, empirically validated solution that improves customer retention and decision-making efficiency.

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 Application-driven integrated framework combining CNN–LSTM-based behavioral modeling with a Dynamic GA for marketing optimization in e-commerce.

 Enhanced LSTM (ELSTM) structure is employed to improve temporal dependency modeling, representing a task-specific refinement tailored for repurchase prediction.

 Dynamic GA (DGA) is utilized to adaptively optimize personalized marketing strategies under cost-efficiency constraints.

 Comprehensive empirical evaluation on a real-world dataset demonstrates performance improvements over multiple classical and Machine Learning (ML) baseline models.

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 The framework provides **practical insights** for integrating predictive analytics with decision-oriented optimization to support intelligent marketing strategies.

2. Relevant Articles

An investigation was conducted to develop creation and transformation mechanisms for e-commerce live streaming adoption [11]. Grounded theory coding was applied to interview data from 42 participants. Findings indicated that customer demand constrained LS usage, while anchors, product messaging, and streaming media influenced attitudes, enhancing engagement and purchase intention. However, limited data restricted generalizability and practical applicability.

Examining the mediation effect of brand trust on customers' purchasing intentions in live streaming e-commerce was undertaken [12]. The method evaluated perceived value across utilitarian, emotional, and social dimensions, along with trust in suppliers and product reliability influencing repurchase intention and revenue perception. Findings highlighted the role of perceived utility. However, regional disparities affecting continued purchasing remained undetermined.

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Forecasting customer purchasing behavior through a data platform built on edge computing technology was addressed [13]. A hybrid model combining Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) was employed. Results indicated that information segmentation improved prediction performance by nearly 2%. However, limited development support and insufficient platform maturity constrained practical implementation.

Developing a modern approach for predicting customer retention in e-commerce retail was addressed [14]. Explainable Artificial Intelligence (XAI), using feature importance measures and partial dependence profiles, identified key retention determinants. Results showed churn propensity was

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influenced by the initial purchase payment value. However, limited resources restricted inclusion of additional predictive models.

Enhancing user satisfaction in e-commerce through Human–Computer Interaction (HCI) technologies was explored [15]. A Deep Neural Network (DNN) predicted consumers' behavioral habits and psychological patterns to generate personalized product recommendations. Experimental results showed high effectiveness, with user satisfaction reaching 93.2%, indicating the system's strong capability in delivering relevant and appealing recommendations.

An influence mechanism model for repurchase behavior among Chinese fast-food consumers was investigated [16]. It employed the Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze Chinese fast-food companies. The investigation found that emotional attachment and self-brand connection positively impact Chinese fast-food repurchase behavior. The investigation had limitations in terms of generalizability.

Forecasting anonymous customer purchase intentions was addressed using a ML model integrating Multi-Behavioral Trendiness (MBT) and Product Popularity (POP) features [17]. The proposed MBT-POP model effectively predicted buying behavior, achieving an F1-score of 0.9031. However, limited incorporation of product reputation information restricted comprehensive analysis and potentially affected overall predictive robustness.

Forecasting customer buying intentions using integrated data analytics and ML techniques was examined [18]. Model performance comparison identified Random Forest (RF) as the most stable classifier, achieving 92.39% accuracy for online shopping prediction. However, advanced approaches such as DL or transfer learning were not explored, and lower efficiency remained a notable limitation.

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Evaluating the usability and security of e-commerce websites [19] using statistical methods and security scanner applications was conducted to examine e-business standards emphasizing e-service quality. Results indicated that the row-wise sum of matrix B formed a (3 × 1) matrix representation. However, the analysis did not encompass all qualifying features of e-commerce websites.

An integrated approach to predicting loss of consumers that combined statistical methods and ML algorithms was assessed in the investigation [20]. The recall of the most effective three learning models measured from 0.56 to 0.72. The investigation's generalizability was limited.

Developing an intelligent prediction model based on a sequential architecture using the Keras DL framework was undertaken [21]. Multiple optimization algorithms were trained on a real-time product market dataset from a cross-border online marketplace to forecast consumer satisfaction. Results demonstrated that the adaptive framework was resilient and stable. However, limitations arose from data source constraints and organization issues.

An Online Food Delivery Service (OFDS) forecasting through the Reptile Search Algorithm with a Hybrid DL-based UX Detection (RSAHDL-UXD) method was suggested [22]. The experimental results showed that the model achieved higher accuracy. Its reliance on specific datasets limited the RSAHDL-UXD model's applicability to other food delivery sites or real-world changing circumstances.

Optimizing e-commerce marketing using DL and data mining was discussed in the research [23]. It established a predictive model for evaluating customer quality based on data mining and the value characteristics of e-commerce agricultural items. Data variation and applicability across diverse rural regions were limited in the research application.

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Estimating Amazon customer feedback using DL techniques, including Latent Dirichlet Allocation (LDA), was explored in [24]. Business validation results showed effective assessment of product requirements under varying demands, supporting product planning and R&D decisions. However, model parameters were insufficiently tuned to specific experimental conditions, limiting optimal performance.

Multilingual big data analysis enabled a multimodal learning framework integrating diverse information sources for cross-border e-commerce [25]. The Adam optimizer proved effective, yielding significant reductions in training loss and improved testing accuracy. However, the approach was not extended to other types of cross-border e-commerce enterprises, limiting broader validation and research expansion.

Modern ML approaches were developed to analyze user activity patterns and build accurate repurchase prediction models [26]. A comprehensive framework integrating collaborative learning techniques with real-time prediction systems enhanced forecasting accuracy. Experimental results demonstrated superior performance over existing methods. However, emerging patterns in customer behavior assessment were not identified, limiting deeper behavioral insights.

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Consumer purchasing decisions were analyzed using the system response concept and perceptual processes, incorporating both cognitive and emotional perspectives [27]. The findings contributed to e-commerce research by explaining how consumers evaluate cognitive and affective factors during purchase decisions. However, cultural and psychological differences limited the applicability of results across different countries.

A service-based logical approach was used to identify components and methods enhancing intentions to use e-commerce platforms by integrating application value into a conceptual model [28]. Findings showed that perceived utility and user satisfaction positively influenced broader platform

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usage intentions. However, reliance on a single platform limited the generalizability of results.

Understanding customers' cognitive and emotional responses to enhance sales of agricultural products in rural e-commerce was addressed using the Customer-Value Hierarchy Model (C-VHM) [29]. Data from users purchasing through a prominent e-commerce platform were analyzed. However, other major platforms, social media channels, and emerging short-form sales avenues were not included.

Improved wind speed forecasting [30] for stable power networks was achieved using a weighted ensemble of MLP, KNR, and LSTM models optimized by the adaptive dynamic grey wolf–dipper throated optimization (ADGWDTO) algorithm. The method achieved Relative Mean Squared Error (RMSE) of 0.0035 and strong feature fitness. However, higher computational complexity and reliance on high-quality time-series data were limitations.

Short-term photovoltaic (PV) power forecasting for grid stability employed a hybrid framework combining Patch Test Set (TST) [31] for temporal feature extraction with SMOA and bSMOA for feature selection and hyperparameter optimization. The model achieved very low MSE and RMSE with high R^2, Willmott Index, and NSE values. However, performance relied on data quality, and scalability required further validation.

Factors influencing student online shopping were identified to enhance prediction accuracy using Support Vector Machine (SVM) classification [32]. Kernel-based SVM was optimized through a GA to determine optimal parameters. The GA-SVM model achieved higher classification accuracy. However, performance depended on kernel selection, and GA optimization increased computational complexity.

Reducing credit risk in agricultural supply chain finance was addressed using a GA-optimized Backpropagation Neural Network (BPNN) [33]. Principal Component Analysis (PCA) selected key assessment indicators,

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and case analysis validated the approach. The GA-BPNN improved convergence, avoided local minima, and enhanced prediction accuracy and speed. However, performance depended on input data quality and required broader contextual validation.

Investigate factors influencing online repurchase intentions to enhance customer retention [34]. The research applied ANN, RF, XGBoost, LightGBM, and LSTM models with preprocessing and feature selection. High predictive accuracy (R^2, low RMSE) identified reviews, ratings, purchasing behavior, and service quality as key factors. Limited sample size and survey-based data may affect generalizability across industries.

2.1 Research Gaps

Recent studies on e-commerce repurchase and churn prediction fall into three categories. First, classical ML models (RF, SVM, and LR) provide interpretability but fail to capture complex temporal patterns in user behavior. Second, DL sequential models (CNN, LSTM, and hybrid CNN–LSTM) improve predictive accuracy by modeling nonlinear and time-dependent interactions but focus mainly on prediction and use static architectures. Third, ensemble or hybrid frameworks enhance robustness but often increase computational cost and lack mechanisms for adaptive marketing decisions.

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In contrast, the proposed DGA-EVMN framework integrates deep behavioral modeling with dynamic marketing optimization. The EVMN captures spatial and temporal patterns, while the DGA adaptively optimizes personalized marketing under changing preferences and cost constraints. Table 1 shows Unlike prior methods, DGA-EVMN not only predicts repurchase probability but also produces actionable, cost-efficient marketing strategies, addressing a key gap in recent DL–based repurchase prediction research.

Table 1: Summary of existing methods, identified limitations, and how the proposed DGA-EVMN framework addresses key gaps

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Ref.	Method / Application Focus	Identified Limitation	Gap Category	How DGA-EVMN Addresses the Gap
[13]	LSTM–CNN with edge computing	Limited scalability and deployment flexibility	Scalability	Centralized DL architecture enables scalable training without edge constraints
[15]	DNN-based HCI recommendation	Poor evaluation of behavioral sequences	Behavioral modeling	EVMN captures temporal dependencies from sequential interaction and transaction data
[16]	Ensemble/fusion models	High computational complexity and training time	Computational efficiency	Vanilla CNN + ELSTM with DGA enables lightweight, efficient learning and decision-making
[17]	MBT-POP for anonymous customers	Missing product reputation and contextual features	Feature incompleteness	Integrates multidimensional behavioral, transactional, and marketing features

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[22]	Hybrid DL for food delivery	Domain-specific datasets limit generalization	Generalizability	Domain-agnostic behavioral framework applicable across e-commerce
[24]	LDA + DL customer feedback	Model parameters insufficiently tuned	Optimization robustness	DGA adaptively optimizes model and marketing parameters
[26]	Feature engineering with ensemble ML	Cannot capture evolving customer behavior	Temporal adaptability	ELSTM models long-term sequential dependencies for adaptive behavior learning

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3. Research Methodology

Developing an innovative DL framework to predict repurchase patterns and improve effective marketing for e-commerce platforms is the objective presented in the research. Predicting e-commerce users repurchase behavior data is obtained from the open-source Kaggle. The preprocessing includes handling missing values through imputation, removing duplicate entries, and applying min-max normalization for obtaining the numerical features. The application of the proposed DGA-EVMN in predicting repurchase behavior and enhancing the marketing for e-commerce users is explored more comprehensively, and the overall processes presented in the research are displayed in Figure 1.

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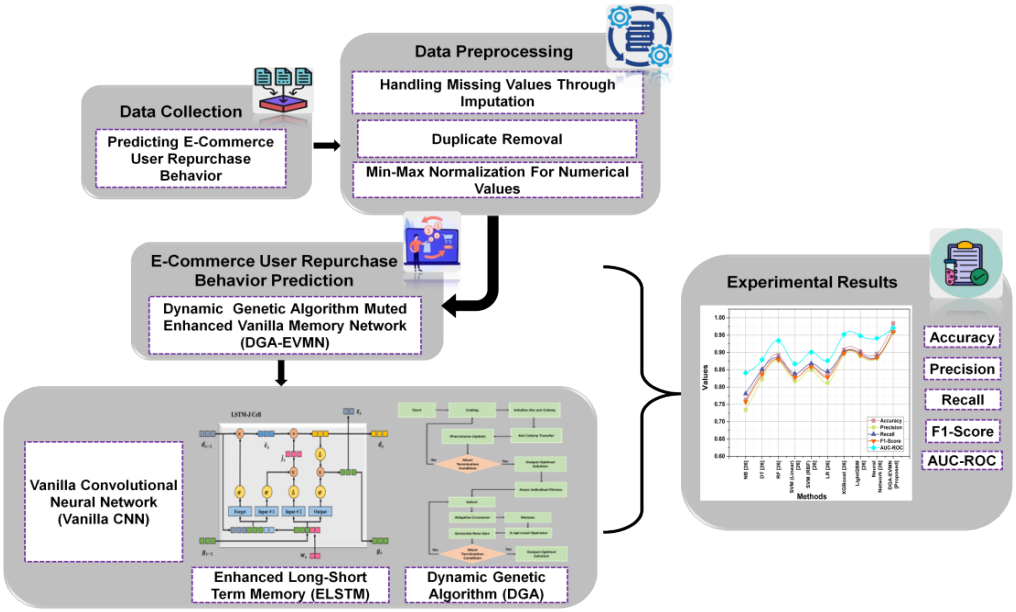

Figure 1: Workflow of Overall Processes Presented in Research

3.1 Data Collection and Preparation

The Predicting e-commerce users' repurchase behavior dataset is obtained from the open source Kaggle https://www.kaggle.com/datasets/programmer3/predicting-e-commerce-user-repurchase-behavior). This dataset includes 1,450 e-commerce users and offers thorough information about their demographics, browsing habits, purchase history, and marketing interactions. Age, gender, geography, income level, device type, buy frequency, average order value, overall time spent on the website, page visits, cart and Wishlist actions, discount received, marketing kinds, and if the user repurchased within 30 days are all important factors considered. The dataset was partitioned into three mutually exclusive subsets: 70% for training, 15% for validation, and 15% for testing. The training set was used for learning model parameters, while the validation set supported hyperparameter tuning and early stopping. The held-out test set was not exposed to the model during training or optimization and was used exclusively for final performance reporting.

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Data preprocessing ensured quality and consistency before modeling. Missing values were statistically imputed, duplicates removed, and numerical features normalized to the [0,1] range using min–max scaling. These steps provided clean, standardized inputs for the DGA-EVMN framework while maintaining focus on the proposed methodology's novel contributions.

A) Handling Missing Values through Imputation: Handling missing values with imputation in an e-commerce platform requires filling the missing data to increase the proposed model's accuracy. Imputation is a technique that uses mathematical techniques to fill missing values in a dataset without altering the overall qualities and values.

B) Removal of Duplicate Entries: Duplicate information is one of the more prevalent and destructive issues for e-commerce businesses. Duplicate data elimination is an important aspect of the process, as it ensures accurate data and prevents issues like inaccurate reporting and adverse customer experiences.

C) Min-Max Normalization: A linear transformation is used in the original data through min-max normalization, including scaled features associated with economy matrices, and it obtains all of the normalized e-commerce information within the range of 0 and 1.

$$X' = \frac{X - X_{Min}}{X_{Max} - X_{Min}} \tag{1}$$

The highest values of financial data and lowest values of financial data are obtained, and every value is replaced through Equation (1). The variable X is the attribute information. Variables X_{Min} and X_{Max} represent the minimum and maximum values. The updated value of each input of data is denoted by X'.

3.2 Predicting Repurchase Behavior through DGA-EVMN

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The ELSTM and DGA components are designed as task-specific adaptations of standard LSTM and GA formulations, aimed at improving temporal modeling and marketing optimization rather than introducing fundamentally new algorithms.

The investigation presents a unique DL framework called DGA-EVMN that integrates the Vanilla CNN (Vanilla CNN) with Enhanced LSTM (ELSTM) that forms as the EVMN approach and DGA to predict repurchase patterns and improve effective marketing in e-commerce platforms.

3.2.1 Enhanced Vanilla Memory Network (EVMN)

Research highlights CNN and LSTM roles within the framework. Vanilla CNN extracts spatial user interaction features, while Enhanced LSTM (ELSTM) fuses previous cell and hidden states to better model temporal dependencies, capturing dynamic repurchase behavior without increasing architectural complexity.

Vanilla CNN: It serves as a basic feature extractor for capturing geographical and interaction-based patterns in e-commerce user data. The output was further smoothed before being received through two thick layers (256 and 128 neurons) to learn complex representations. Finally, a softmax output layer calculates the probability of repurchase. Its hidden layers (h_0, h_1, and h_2) enhance learned features using nonlinear transformations. The back propagation optimizer enables adaptive weight adjustment. Hyperparameters of the Vanilla CNN are determined in Table 2.

Table 2: Hyperparameters of Vanilla CNN

Hyperparameter	Final Value
Number of Convolutional Layers	4
Filter Sizes	(3 × 3)
Number of Filters	{32, 64, 128, 256}
Pooling Type	Max Pooling
Activation Function	ReLU
Dropout Rate	0.3

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Dense Layer Units	{256, 128}
Optimizer	Adam

ELSTM: LSTMs were forms of RNN capable of developing long-term dependencies. It is intended to avoid dissolving or exploding gradients. LSTM for prediction and classification functions effectively with time-series data. LSTM units have many topologies and typically consist of an input, forget, memory cell, and output gates. Each LSTM cell produces new hidden state and cell state data. The computation for an LSTM cell is provided (Equations 2-7).

$$Fgt_s = \sigma_h(Z_{Fgt}a_s + V_{Fgt}g_{s-1} + y_{Fgt}) \tag{2}$$

$$In_s = \sigma_h(Z_{In}a_s + V_{In}g_{s-1} + y_{In}) \tag{3}$$

$$Out_s = \sigma_h(Z_{Out}a_s + V_{Out}g_{s-1} + y_{Out}) \tag{4}$$

$$\hat{d}_s = \sigma_d(Z_d a_s + V_d g_{s-1} + y_d) \tag{5}$$

$$d_s = Fgt_s \circ d_{s-1} + In_s \circ \hat{d}_s \tag{6}$$

$$g_s = Out_s \circ \sigma_g(d_s) \tag{7}$$

In this instance, $a_s$represents the input vector to the LSTM unit, whereas In_s, Fgt_s,Out_s and $\hat{d}_s$denote the input, forget, output gates, and cell unit (g_s) of activation vectors. Variable d_s is the cell state vector. The weight matrices are denoted as Z and V, whereas y is the metric of the bias vector. Variable

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σ_h is the tangent sigmoid function, whereas σ_d and σ_g are hyperbolic tangential functions.

In traditional LSTM cells, the present cell state d_s is important for short-term memory and directly affects the hidden state and final output. However, the previous state d_{s-1} is only associated with the late concatenation of forget and input gates. The research develops a new traveling approach to cell state with tangent functions (σ and Δ), as illustrated in Figure 2. In the ELSTM cell, previous state d_{s-1} collaborates with previous hidden state g_{s-1} at the forget gate, along with current input a_s gate #1.

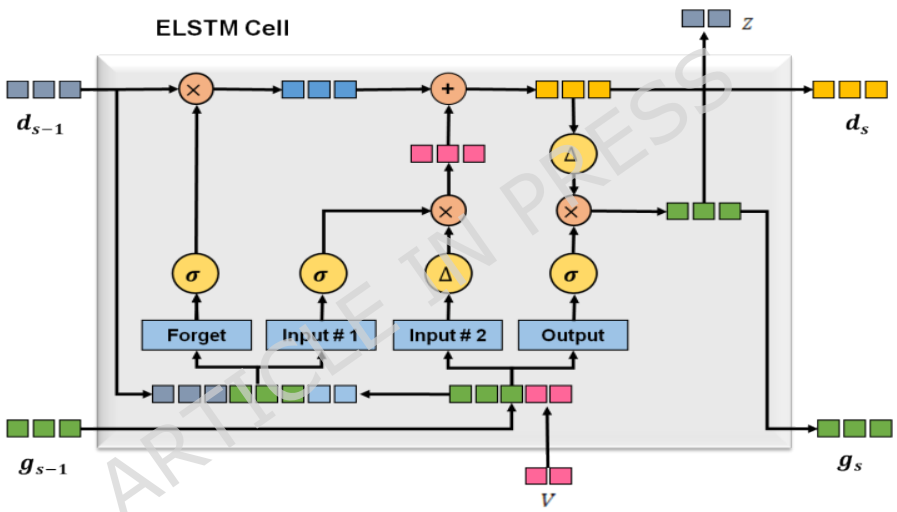

Figure 2: Structure of the ELSTM Cell

The ELSTM cell allows for joint concatenation of d_{s-1} and a_s; in the forget gate of an ELSTM cell, the output $\overline{Fgt}_s$ is determined by early conjunction of a_s,g_{s-1}, and d_{s-1}, which is computed using Equation (8).

$$\overline{Fgt}_s = \sigma\left(Z^e_{ag} \times a_s + Z^e_{gg} \times g_{s-1} + Z^e_{dg} \times d_{s-1} + y_{\overline{Fgt}_s}\right) \tag{8}$$

The weight (Z^e_{ag}) from the cell to the hidden state determines the information filtering from both the long and short term memory, which is

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denoted by Z_{dg}^{e} and Z_{gg}^{e}, while the bias is represented by $y_{\overline{Fgt}_s}$. Input gate #1 of an ELSTM cell, $\overline{In}_{1s}$ is determined by the early conjunction of a_s and g_{s-1}, as shown in Equation (9).

$$\overline{In}_{1s} = \sigma\left(Z_{ag}^{1} \times a_s + Z_{gg}^{1} \times g_{s-1} + Z_{dg}^{1} \times d_{s-1} + y_1\right) \tag{9}$$

Variable $\overline{In}_{1s}$ represents input information from both long-term and short-term memory, while $\overline{In}_{2s}$ is based solely on short-term memory. ELSTM provides a joint concatenation of cell state by keeping input gate #2 and output gate unchanged, similar to ELSTM's late concatenation. The ELSTM hyperparameters are determined in Table 3, and the EVMN algorithm is shown in Algorithm 1.

Table 3: Hyper-parameters of EVMN Approach

Hyperparameter	**Ranges**
Input Dimension	128
Batch Size	64
Learning Rate	$0.001 - 0.0001$
Dropout Rate	0.3
LSTM Units	128
Number of LSTM Layers	$1 - 3$
Sequence Length	30
Epochs	100
Regularization	L2 Regularization ($\lambda = 0.001$)
Early Stopping Patience	10

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Hyperparameters of the Vanilla CNN and ELSTM were tuned using grid search with validation performance. Multiple configurations were evaluated, optimizing F1-score and AUC-ROC while controlling overfitting through early stopping. The best-performing settings are reported in Tables 1 and 2.

All hyperparameters, including LSTM units, sequence length, batch size, learning rate, dropout, and DGA parameters, were optimized via grid search on validation data. Final values in Tables 1–3 ensure reproducibility and optimal performance.

Algorithm 1: *EVMN*

Input*: preprocessed e-commerce user data*

Def Vanilla CNN for spatial feature extraction

cnn_features = VanillaCNN (input_data, num_layers=3, num_filters=[32, 64, 128], filter_size=(3,3), pooling='max', dropout_rate=0.3)

Def ELSTM for temporal dependency modeling

lstm_output = ELSTM(cnn_features, units=128, num_layers=2, sequence_length=30, dropout_rate=0.3)

repurchase_prob = Dense(lstm_output, units=1, activation='sigmoid')

model.fit(input_data, target_labels, batch_size=64, epochs=100, validation_split=0.2)

prediction = model.predict(new_user_data)

Output*: predicted repurchase probability*

3.2.2 Dynamic Genetic Algorithm (DGA)

GA is a computational search mechanism that simulates the process of natural selection to discover optimal or near-optimum responses to difficult optimization and searches, particularly issues with large or unclear search

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spaces in predicting e-commerce users repurchase behaviors. Traditional GA in marketing approach optimization frequently experiences issues such as premature convergence and constrained local search capabilities.

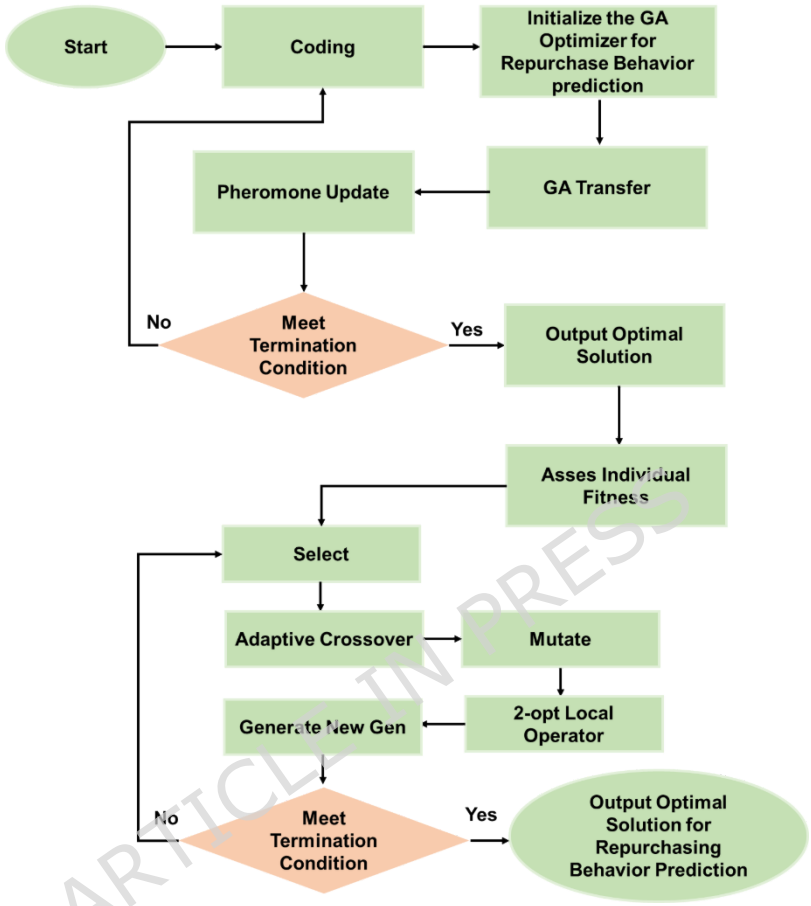

Figure 3: Algorithmic Flow of DGA Approach

Figure 3 presents the DGA-EVMN workflow, where EVMN predicts repurchase probabilities and DGA optimizes personalized marketing through adaptive crossover and mutation for cost-efficient decisions.

Individual Encoding and the Fitness Function: In the DGA method, coding processes are used to map phenotypes to genotypes. A sequence-based encoding approach is used to describe customer behavior and methods of marketing, with each user's activities like 1,2,…,j. The identified characters are then combined into sequences that represent the temporal sequence of interactions.

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The DGA fitness function balances repurchase probability, engagement, and CLV while penalizing excessive marketing costs for optimized decisions. The resulting scalarized multi-objective fitness function is defined as:

$$\text{Fitness} = \alpha \bullet p_{repurchase} + \beta \bullet \text{Engagement} + \gamma \bullet \text{CLV} - \delta \bullet \text{cost} \quad (10)$$

Equation (10) represents where $p_{repurchase}$ represents the predicted repurchase probability from the EVMN model, *Engagement* captures user interaction intensity, CLV estimates long-term customer value, and *Cost* denotes marketing expenditure. The weighting coefficients α, β, γ, and δ control the trade-off among objectives and are empirically selected to ensure balanced optimization. This formulation enables the DGA to approximate Pareto-optimal solutions while maintaining computational efficiency.

DGA Operators: In e-commerce, the DGA is used to optimize customized marketing techniques. Iterative selection, crossover, and mutation are used to generate potential solutions such as savings, incentives, and suggestions.

Initialization: The low average quality of population individuals is attributed to the poor fitness of GA initiated individuals and the algorithm's limited convergence capacity. As a result, the IGA maintains high fitness at the start, allowing for fast convergence. The algorithm steps are provided below.

Phase 1 - Initialize the algorithm's configurations. The population size, maximum number of repetitions, pheromone strength, information heuristic factor, and expectation heuristic factor have to be established.

Phase 2 – Resolving the traditional GA solution by DGA.

- Analyze user interactions and transactions to determine the distance (c_{ji}) between customers (j) and (i) based on behavior patterns.

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- Initialize each visited-history record, and then let each one choose the next user or action based on the transition probability.
- Update the global pheromone matrix to strengthen high-quality behavioral patterns that lead to stronger repurchase predictions.
- Record each current solution, update the optimum sequence discovered previously, then transmit data to initialize the DGA population.

Phase 3 – The output results are utilized as the initial population for the traditional DGA performance.

Selection: Using the anticipated value approach permits addressing the issue with traditional methods. The primary concept is as follows.

- Determine the number of people in a population fitness ($Fit_{Average}$) that possess each chromosome to passed down through successive generations. The estimation formula is shown in Equation (11).

$$m_j = \frac{t_j}{Fit_{Average}} = \frac{t_{j*N}}{\sum_{j=1}^{N} t_j} \tag{11}$$

Variable t_j indicates an individual's fitness value of $j = 1,2,\ldots,N$. Variable N is the individual's overall quantity and $Fit_{Average}$ is the average fitness value.

- If a chromosomal individual is chosen for the next-stage crossover procedure, its m_j in the following generation is reduced by 0.5. If not chosen, m_j is reduced by one.
- Individuals with an anticipated survival value (m_j) less than zero will be excluded from the selection process.

Crossover: It employs the sequential crossover operator to finish the crossover process. In the early stages of GA operations, there is a significant population disparity. The formula for computing a dynamic crossover probability is formulated as Equation (12).

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$$H_d = \begin{cases} \frac{d_1(t_{maximum} - t)}{t_{maximum} - t_{Average}}, & t \geq t_{Average} \\ d_2, & e < Fit_{Average} \end{cases}$$

(12)

Where, $t_{maximum}$ denotes the group's highest fitness amount. The average fitness value for a group is denoted as $t_{Average}$. The higher fitness level of the two individuals engaged in the cross is denoted by H_d, which is generally constant. Variables d_1 and d_2 fall between 0 and 1.

Mutation: It employs the reverse mutation operator to finish the mutation process. Reverse mutation involves randomly selecting an interception location on a chromosomal coding string and then reversing the collected gene sequence based on the reversal probability. A reverse mutation is a specific type of basic mutation.

Dynamic crossover and mutation probabilities in the DGA are computed based on individual fitness relative to the best and average fitness, as detailed in Algorithm 2 and Equation (12). This ensures higher-fitness individuals preserve quality solutions, while lower-fitness individuals explore new possibilities.

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Objective Function and optimization strategy

The DGA is designed to jointly optimize repurchase prediction effectiveness and marketing cost efficiency. Specifically, the optimization objective balances the predicted repurchase probability generated by the EVMN model with the cost associated with marketing interventions. The fitness function is defined as: The optimization objective is defined as:

$$Fitness = \alpha \bullet P_{repurchase} - \beta \bullet C_{marketing}$$

(13)

Equation (13) shows, where $P_{repurchase}$ represents the predicted probability of a user repurchasing within 30 days, and $C_{marketing}$ denotes the

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normalized cost of the selected marketing strategy. The weighting coefficients α and β control the trade-off between maximizing customer retention and minimizing marketing expenditure.

Marketing cost reflects intervention value per user. Repurchase probabilities guide strategies: high-probability users receive low-cost personalization, moderate users get incentives, and low-probability users receive minimal intervention, ensuring cost-efficient decisions aligned with predicted behavior. Table 4 demonstrates hyperparameters used in the DGA approach. Algorithm 2 determines the DGA algorithm.

Table 4: Hyperparameters Utilized in DGA

Hyperparameter	**Values**
Population Size	50 − 200
Maximum Generations	100 − 500
Crossover Probability	Dynamic: 0.6 − 0.9
Mutation Probability	Dynamic: 0.01 − 0
Pheromone Update Factor	0.1 − 1.0
Heuristic Factor	0.5 − 1.0
Expectation Factor	0.5 − 1.0

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DGA hyperparameters were empirically optimized using convergence trends, fitness improvement, and solution stability. Candidate configurations were tested, balancing exploration and exploitation. Final settings, shown in Table 4, consistently enhanced repurchase prediction accuracy and marketing cost efficiency.

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A single-objective GA ensures stable convergence, lower computational complexity, and integration with EVMN outputs. Multiple objectives, including repurchase likelihood and marketing cost, are unified into a scalar fitness function, avoiding Pareto instability and enabling efficient optimization for moderate datasets and practical deployment.

Rationale for Selecting EVMN over Alternative Architectures

Repurchase prediction in e-commerce requires learning from heterogeneous behavioral data containing localized interaction patterns and evolving temporal dependencies. The proposed EVMN integrates Vanilla CNN and ELSTM to address these aspects. CNN captures spatial correlations among multidimensional user interactions, identifying short-range dependencies, while ELSTM models temporal dynamics, overcoming limitations of traditional LSTM architectures applied directly to raw behavioral data.

Compared to standard LSTM, EVMN separates spatial feature extraction and temporal modeling. CNN converts raw behavioral inputs into structured feature maps, enabling ELSTM to capture long-term dependencies effectively. This hybrid design improves predictive stability, reduces noise sensitivity, and enhances repurchase forecasting accuracy.

Transformer models perform well on large datasets but require high computation. For medium-sized e-commerce data, they risk overfitting and reduced interpretability.

Therefore, the EVMN hybrid architecture provides a balanced solution for repurchase prediction, combining strong modeling capability, computational efficiency, interpretability, and practical suitability for adaptive e-commerce marketing deployment.

Algorithm 2: Dynamic Probability Update in DGA

Input: Population P, fitness values f_i, P_c^max, P_m^max

Output: Updated population with adaptive crossover and mutation

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```
For each generation g do
    Evaluate fitness f_i for all individuals
    Compute f_max and f_avg
    For each selected individual i do
        If f_i ≥ f_avg then
            Pc = Pc_max × (f_max − f_i) / (f_max − f_avg)
            Pm = Pm_max × (f_max − f_i) / (f_max − f_avg)
        Else
            Pc = Pc_max
            Pm = Pm_max
        End If
        Apply crossover with probability Pc
        Apply reverse mutation with probability Pm
    End For
    Update population using elitism
End For
```

An integrated algorithm 3 DGA-EVMN combines the DL evolutionary computing to improve e-commerce repurchase behavior prediction and marketing optimization. To capture spatial and temporal links in user data, the EVMN component incorporates Vanilla CNN and ELSTM networks. The hybrid memory network provides superior interpretability, decreased computational demands, and improved prediction accuracy.

Algorithm 3: *DGA-EVMN*

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Input: *Preprocessed e-commerce data X, labels Y*

cnn_features = VanillaCNN(X, layers=4, filters=[32,64,128,256], pool='max')

lstm_output = ELSTM(cnn_features, units=128, layers=2, seq_len=30)

repurchase_prob = Dense(lstm_output, units=1, activation='sigmoid')

model.fit(X, Y, batch=64, epochs=100, val_split=0.2, early_stopping=10)

Initialize population P of marketing strategies

For *each p in P: fitness[p] = model.predict(p) / cost(p)*

For *gen = 1 to MaxGen:*

Select top individuals based on fitness

Perform dynamic crossover and reverse mutation

Evaluate offspring: fitness = predict()/cost()

Update the population with the best individuals

best_strategy = argmax(fitness)

Return repurchase_prob, best_strategy

Output: *Predicted repurchase probability & optimized marketing strategy*

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4. Results and Discussions

The research intends to determine the DL framework called DGA-EVMN for efficient repurchase behavior forecasting and enhancing the marketing for e-commerce customers. There are several sections, such as evaluation criteria, comparison phases, and discussions, provided below that show the DGA-EVMN functions. The implementation through several system configurations is shown in Table 5.

Table 5: System Configurations

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Components	Details
Programming Language	Python
Operating System (OS)	Windows 11
RAM	32 GB
Processor	Intel Core i7

A comprehensive evaluation of the proposed DGA-EVMN system was conducted using multiple performance metrics to assess repurchase prediction accuracy and marketing optimization effectiveness. Indicators such as conversion rate, marketing cost efficiency, and resource utilization measured the practical impact and overall efficiency of optimized advertising strategies.

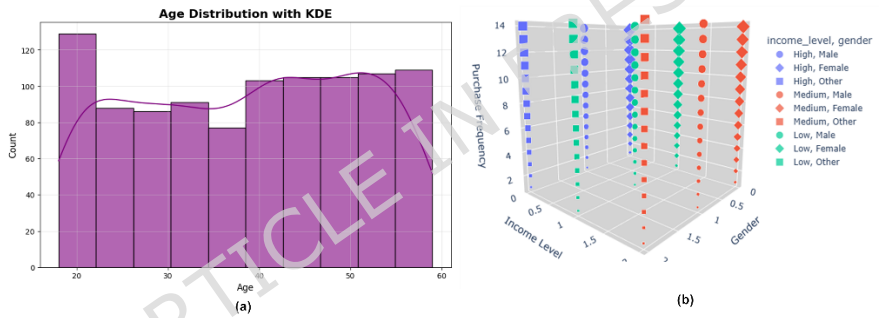

Figure 4: Evaluation of (a) Age Distribution using Kernel Density Estimation (KDE) and (b)Purchase Frequency by Gender

Figure 4 (a) illustrates the age distribution using KDE, showing a balanced sample of users aged 18–60, with slight peaks in the mid-20s and late 50s indicating higher engagement. **Figure 4(b)** shows purchase frequency by gender, with similar median values and only minor variations across categories.

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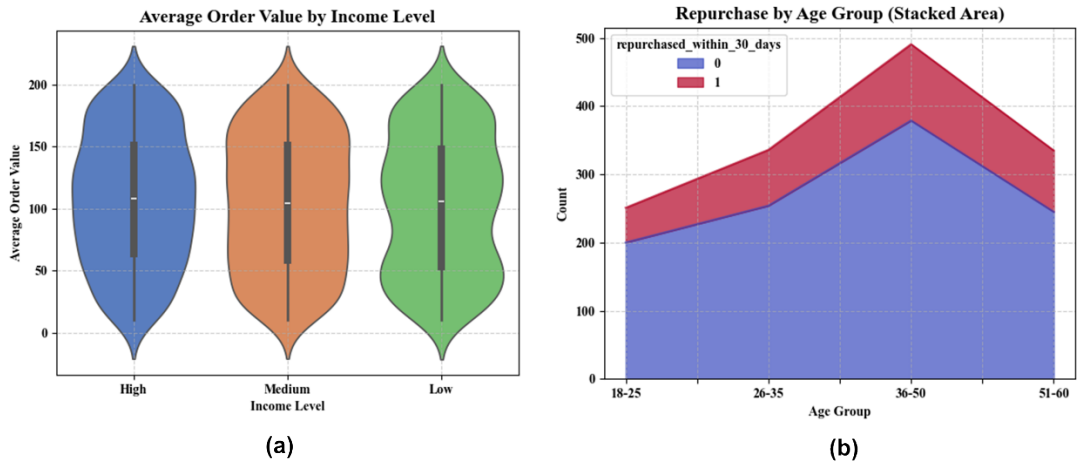

Figure 5: Visual Depiction of (a) Order Value with Income Level and (b) Repurchasing with Age Group

Figure 5(a) presents average order value across income levels, showing similar purchasing patterns among groups, with slightly higher spending variability observed among high-income users. **Figure 5(b)** shows 30-day repurchase rates rising from ages 18–25 to 36–50, peaking among middle-aged users, then declining at 51–60, indicating targeted marketing here can improve Return on Investment.

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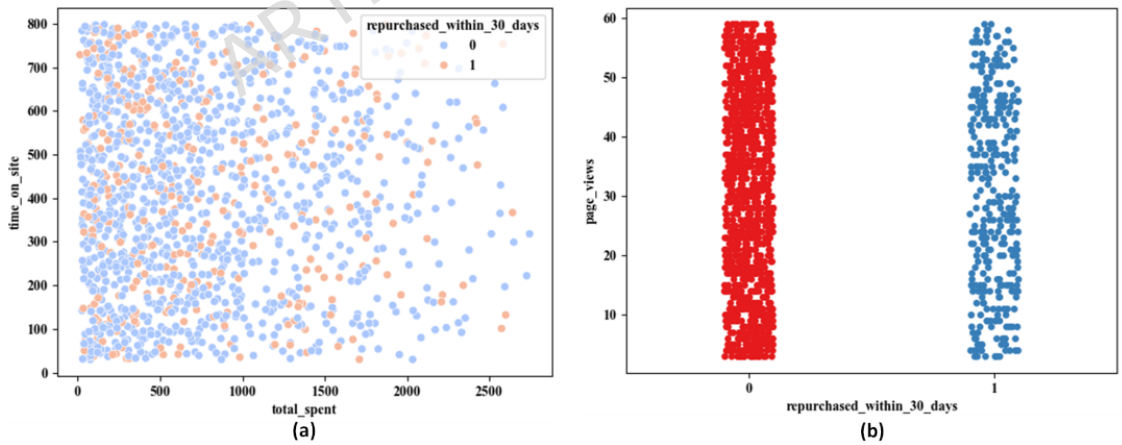

Figure 6: Representation of (a) Total Spending and Time on Site and (b) Comparison of Page Views and Repurchase

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Figure 6(a) illustrates the relationship between total expenditure and time spent on the site, categorized by items repurchased within 30 days, highlighting repurchase behavior trends. **Figure 6(b)** shows that repurchased users (label 1) have higher page views than non-repurchasers (label 0), indicating frequent browsing increases purchase likelihood and supports DL-based predictive marketing approaches.

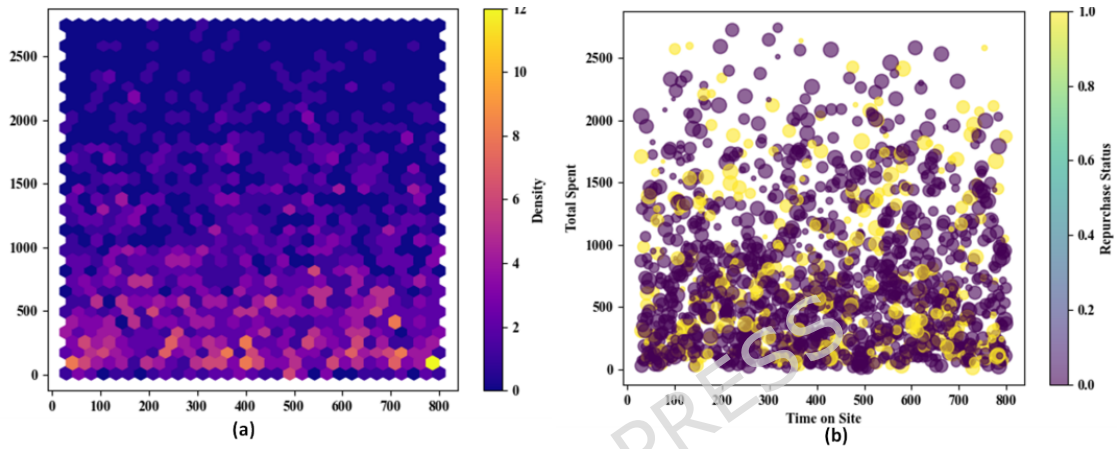

Figure 7: Estimation Outcomes of (a) Time on Site versus Total Spent Graph and (b) Time versus Total Spent

Figure 7(a) illustrates time on site versus total spending, using visual density to highlight regions with high user activity and engagement patterns. **Figure 7(b)** presents time versus total spent, with bubble size showing page visits and color indicating repurchase. Larger, vibrant bubbles reflect high engagement, strongly correlating with repurchase probability and validating these features.

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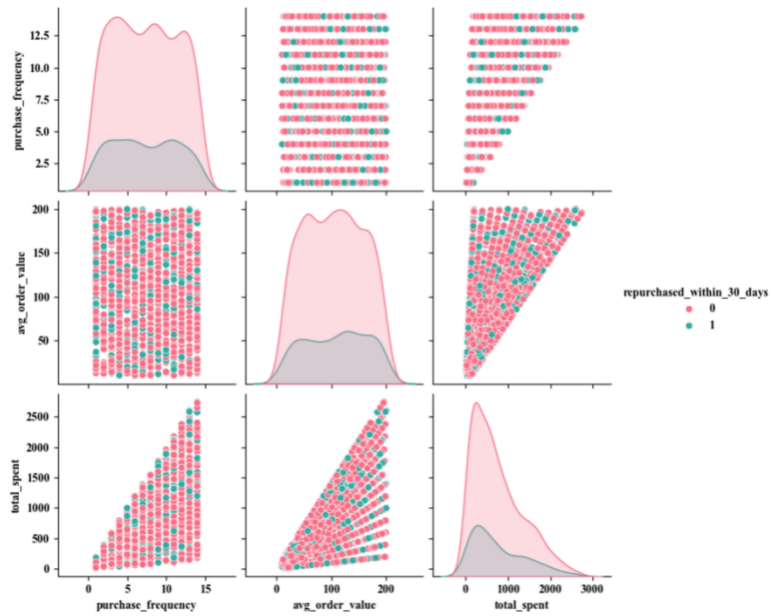

Figure 8: Visual Representation of the Relationship among Three Key Features in Repurchasing Behavior

Figure 8 illustrates the relationships between purchase frequency, average order value, and total spending with 30-day repurchase status. Diagonal plots show feature distributions for repurchased (1) and non-repurchased (0) users, highlighting behavioral differences and indicating these features are reliable predictors of repurchase behavior.

Table 6 represents the outcomes of repurchase and non-repurchase values. There are 1450 e-commerce users determined in the research from the user; 23.31% of users are provided as repurchased users, and the other 79.69% are identified as non-purchased users. The findings show efficiency in exploring the rate of repurchasing and non-repurchasing.

Table 6: Outcomes of Repurchase and Non-Repurchase

Metric	Value
Total Users	1,450
Repurchase Rate (%)	23.31%

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Non−repurchase Rate (%	76.69%

4.1 Comparison Phase

An innovative DGA-EVMN model is proposed in the research to effectively forecast and improve the repurchasing behavior of the e-commerce customer. In this section, a comparison between the proposed DGA-EVMN and various existing techniques, which include Naive Bayes (NB) [26], Decision Tree (DT) [26], RF [26], Support Vector Machine Linear (SVM (Linear)) [26], SVM Radial Basis Function (SVM (RBF)) [26], Logistic Regression (LR) [26], Extreme Gradient Boosting (XGBoost) [26], Light Gradient Boosting Machine (LightGBM) [26] and Neural Network [26], Artificial Neural Network (ANN) [34], Random Forest Classifier (RFC) [34], LightGBM [34], XGBoost [34] is performed. The accuracy, precision, recall, Area under the Receiver Operating Characteristic Curve (AUC-ROC), and F1-score for classification. Table 7 shows the equations and explanations of the performance metrics.

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Table 7: Equations and Explanations of the Performance Metrics

Metrics	Explanations	Equations
AUC-ROC	It determines the possibility that a positive instance selected at random that provided greater importance than an adverse instance selected at random.	$\text{AUC}-\text{ROC} = \int_0^1 \text{TPR}(\text{FI}$
Accuracy	It is a percentage of accurate results of both real positives and real negatives among all test results.	$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$
Precision	It determines the percentage of actual positive forecasting on the entire positive forecast.	$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$

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Recall	The ability of a model to identify each accurate positive case in a dataset is measured by a metric called recall.	$\text{Recall} = \frac{TP}{TP + FN}$
F1-score	A single ML evaluation metric called the F1-score provides a balanced assessment of a classification model's performance by representing a harmonic average of precision and recall.	$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Reca}}{\text{Precision} + \text{Reca}}$

Note: TP – True Positives, TN – True Negatives, FP – False Positives, and FN – False Negatives, TPR - True Positive Rate, FPR – False Positive Rate, X - Data Point and dX - Output Function.

- **Comparative Performance Evaluation of Baseline Models and Proposed DGA-EVMN on the E-Commerce Behavioral Dataset**

The performance comparison was conducted using the large-scale e-commerce behavioral dataset described in [26], comprising nearly 2.3 million user sessions, 847,000 customers, and 156,000 products across 23 categories. The dataset captures diverse transactional and browsing behaviors, providing a reliable foundation for predictive analytics. The proposed model was trained on existing large-scale dataset. All baseline methods and the proposed DGA-EVMN were trained under existing data, as shown in Table 8 and Table 9.

Table 8: Proposed model was trained on a large-scale existing e-commerce behavioral dataset

Methods	**Metrics**		
	Precision	**Recall**	**AUC-ROC**
NB [26]	0.734	0.781	0.841

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DT [26]	0.823	0.851	0.879
RF [26]	0.876	0.883	0.934
SVM (Linear) [26]	0.817	0.839	0.867
SVM (RBF) [26]	0.849	0.868	0.901
LR [26]	0.812	0.845	0.876
XGBoost [26]	0.895	0.901	0.952
LightGBM [26]	0.889	0.897	0.948
Neural Network [26]	0.883	0.887	0.941
DGA-EVMN [Proposed]	0.921	0.932	0.953

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Table 9: Proposed method trained on the [29] existing dataset

Method	**Accuracy**	**F1 score**
NB [26]	0.763	0.757
DT [26]	0.847	0.837
RF [26]	0.892	0.879
SVM (Linear) [26]	0.829	0.828
SVM (RBF) [26]	0.861	0.858
LR [26]	0.834	0.828
XGBoost [26]	0.908	0.898

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LightGBM [26]	0.903	0.893
Neural Network [26]	0.896	0.885
DGA-EVMN [Proposed]	0.953	0.957

- **Comparison Between ANN and Proposed DGA-EVMN Using Online Consumer Survey Data**

The experimental analysis was conducted using the online consumer survey dataset reported in [34], consisting of responses from 438 participants. The proposed DGA-EVMN was trained and evaluated on this existing dataset (shown in Table 10) under consistent experimental settings.

Table 10: The proposed DGA-EVMN was trained under online consumer survey dataset [34]

Method	Accuracy	F1-Score	MAE
ANN [34]	0.923	0.94	0.141
DGA-EVMN [Proposed]	0.958	0.959	0.139

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- Comparison of Baseline Models and Proposed DGA-EVMN on the real-world e-commerce datasets

The comparative evaluation used the real-world e-commerce dataset containing structured e-commerce behavioral features for repurchase prediction. Both baseline models from and the proposed DGA-EVMN were trained and tested dataset to ensure fairness. Results (Table 11, 12 and Figure 9) indicate that DGA-EVMN delivers superior performance while compared existing dataset, demonstrating improved predictive capability.

Table 11: Performance comparison of existing methods and DGA-EVMN trained on the Proposed dataset

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Methods	Metrics		
	Precision	**Recall**	**AUC-ROC**
NB	0.743	0.784	0.847
DT	0.832	0.856	0.882
RF	0.867	0.886	0.936
SVM (Linear)	0.871	0.843	0.869
SVM (RBF)	0.894	0.870	0.905
LR	0.823	0.848	0.878
XGBoost	0.897	0.908	0.956
LightGBM	0.878	0.898	0.949
Neural Network	0.856	0.888	0.944
DGA-EVMN [Proposed]	0.958	0.962	0.971

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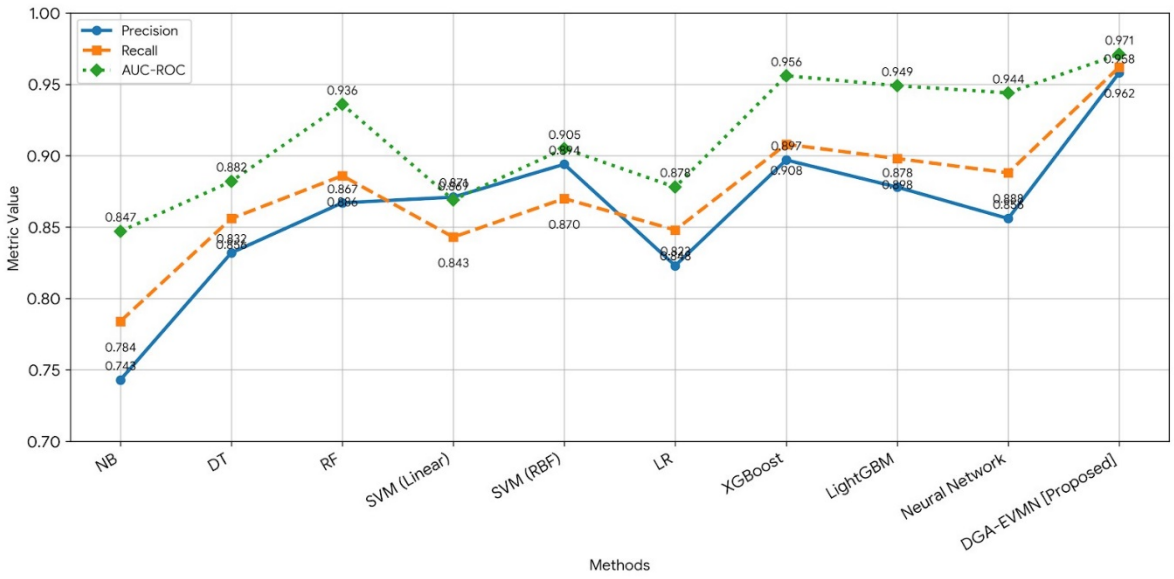

Figure 9: Visual Depiction of Comparison Results

Table 12: Analysis of existing models and proposed DGA-EVMN on the proposed e-commerce behavioral dataset.

Method	Accuracy	F1 score
NB	0.764	0.758
DT	0.848	0.838
RF	0.894	0.883
SVM (Linear)	0.832	0.831
SVM (RBF)	0.863	0.863
LR	0.836	0.835
XGBoost	0.909	0.901
LightGBM	0.905	0.896
Neural Network	0.898	0.887
DGA-EVMN [Proposed]	0.984	0.960

- **Performance Comparison of ANN and Proposed DGA-EVMN Based on Suggested metrics**

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The performance evaluation utilized the real-world e-commerce dataset [Proposed] with structured consumer behavioral features. Both ANN and the proposed DGA-EVMN were trained under identical conditions to ensure fairness. Results show that DGA-EVMN achieves higher accuracy and F1-score with lower MAE (Table 13), indicating enhanced predictive performance.

Table 13: comparison between ANN and DGA-EVMN trained on the Proposed dataset

Method	Accuracy	F1-Score	MAE
ANN	0.925	0.95	0.140
DGA-EVMN [Proposed]	0.984	0.960	0.121

5-Fold Cross Validation

5-Fold Cross Validation was employed to rigorously evaluate the DGA-EVMN model's predictive performance for e-commerce repurchase behavior. The dataset is partitioned into five equal folds (Table 14), iteratively using four for training and one for testing. Averaging performance across all folds provides a robust, unbiased estimate of accuracy, ensuring the model reliably captures complex user interaction patterns and generalizes effectively to unseen data.

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Table 14. 5-Fold Cross-Validation Performance of DGA-EVMN

Fold	Accuracy (%)	F1-score	Precision	Recall	AUC-ROC
Fold 1	98.2	0.958	0.956	0.960	0.912
Fold 2	98.5	0.961	0.959	0.963	0.917
Fold 3	98.3	0.959	0.957	0.961	0.921

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Fold 4	98.6	0.962	0.960	0.964	0.919
Fold 5	98.4	0.960	0.958	0.962	0.923
Mean ± 95% CI	**98.4 ± 0.15**	**0.960 ± 0.002**	**0.958 ± 0.002**	**0.962 ± 0.002**	**0.971± 0.006**

Table 14 illustrates the 5-fold evaluation of DGA-EVMN shows Accuracy 98.2–98.6%, F1-score 0.958–0.962, Precision 0.956–0.960, Recall 0.960–0.964, and AUC-ROC 0.971± 0.006. The mean ± 95% CI indicates Accuracy 98.4 ± 0.15%, F1-score 0.960 ± 0.002, Precision 0.958 ± 0.002, Recall 0.962 ± 0.002, demonstrating consistent and reliable model performance.

Paired t-test

A paired t-test was applied to evaluate the statistical significance of performance differences between the DGA-EVMN and baseline models for repurchase prediction. By comparing paired predictions on the same dataset, it quantifies whether observed improvements in metrics, such as accuracy or F1-score, are statistically significant, providing t-values and p-values to validate the model's enhanced predictive capability.

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Table 15: Paired t-test comparison of EVMN and DGA-EVMN showing mean accuracy, variability, and statistical significance

Model	**Mean (%)**	**SD**	**t-value**	**p-value**	**Significance**
EVMN	96.9	0.22	14.86	< 0.001	Significant
DGA-EVMN (Proposed)	98.4	0.15	28.73	< 0.001	Significant

Table 15 compares EVMN and the proposed DGA-EVMN using a paired t-test. DGA-EVMN achieves a higher **Mean 98.4%** with lower SD (0.15)

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compared to EVMN (96.9%, 0.22), with **t-value 28.73** and **p-value <0.001**, indicating a statistically significant improvement. These results demonstrate DGA-EVMN's superior predictive performance.

Table 16: Standardized Hyperparameter Optimization and Evaluation Protocol for Baseline Models

Model	Optimization Strategy	Main Tuned Parameters	Validation Method	Dataset Split Consistency
NB [26]	Grid Search	Smoothing factor (α)	5-fold CV	Yes
LR [26]	Grid Search	Learning rate, Regularization (C)	5-fold CV	Yes
DT [26]	Grid Search	Max depth, Min samples split, Criterion	5-fold CV	Yes
RF [26]	Grid Search	No. of trees, Max depth, Feature subset	5-fold CV	Yes
SVM (Linear/RBF) [26]	Grid Search	C, Kernel width (γ)	5-fold CV	Yes
XGBoost [26]	Grid Search	Learning rate, Depth, Estimators, Subsample	5-fold CV	Yes
LightGBM [26]	Grid Search	Leaves, learning rate, Boosting rounds	5-fold CV	Yes
Neural Network [26]	Grid + Random Search	Hidden layers, neurons per layer, learning rate, batch size, epochs	5-fold CV	Yes
ANN [34]	Grid + Random Search	Hidden layers, Neurons, Batch size, LR	5-fold CV	Yes

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Table 16 summarizes the unified hyperparameter tuning strategy and evaluation settings applied uniformly to all baseline models. This

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standardized protocol ensures fairness, reproducibility, and unbiased performance comparison across different learning approaches.

4.2 Discussions

The research developed a novel DGA-EVMN method for the prediction and enhancement of the repurchasing behavior of e-commerce users. A comparison among the proposed and traditional techniques demonstrated the efficiency in forecasting the repurchase behavior and marketing enhancement of the e-commerce customers. There were several limitations identified in the traditional methods. Most existing predictive models, including NB [26], DT [26], RF [26], and SVM (Linear and RBF) [26], have limited ability to capture the nonlinear and dynamic behavioral patterns of e-commerce consumers. Although LR [26] and XGBoost [26] managed the high-dimensional data, they frequently show feature independence and are unable to adapt to changing user preferences. The LightGBM [26] increases computing efficiency and still has issues with hyperparameter selection and data imbalance. While Neural Networks [26] ANN [34] RFC [34], LightGBM [34], XGBoost [34] were effective for complex pattern learning, they frequently over-fit small or noisy datasets and lack optimization mechanisms for marketing decision-making.

Overall, the research focused on prediction accuracy over the integration of behavioral dynamics and marketing optimization. As a result, it cannot deliver adaptive, cost-effective, and targeted marketing interventions that maximize client retention and repurchase possibility in competitive e-commerce markets. The proposed DGA-EVMN method effectively resolved the existing technique’s limitations by combining DL's temporal modeling with DGAs' adaptive search capabilities. The effectiveness of the proposed framework stems not from the introduction of new standalone algorithms,

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but from the coordinated integration of predictive modeling and adaptive optimization, enabling actionable marketing decisions in practical e-commerce environments. The research utilized a single publicly available dataset that provides comprehensive behavioral, transactional, and marketing interaction features, which are sufficient to support reliable model training and evaluation for the intended predictive task. To further ensure robustness despite the moderate dataset size, rigorous validation strategies such as five-fold cross-validation, confidence interval reporting, paired statistical testing, and validation-based hyperparameter tuning were implemented to reduce overfitting risk and enhance result stability. Future work focuses on extending the framework to multiple large-scale datasets and conducting cross-dataset benchmarking to further validate generalizability and strengthen external reliability.

Practical implication: The proposed DGA-EVMN framework can significantly improve e-commerce decision-making by accurately predicting repurchase behavior, enabling targeted marketing, optimizing resource allocation, and enhancing customer retention, ultimately boosting revenue and operational efficiency.

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4.3 Limitations and Future Scopes

The DGA-EVMN framework demonstrates strong predictive performance but introduces moderate computational overhead. On an Intel i7 setup, EVMN training required 18–25 minutes, with the DGA increasing optimization time by 30–40%. Larger datasets may require GPU acceleration to reduce convergence time, and current imputation is limited to standard techniques.

Future work will focus on GPU-based training, parallelized GA operations, mini-batch evolutionary strategies, and distributed learning to improve scalability for large-scale e-commerce environments. Additionally, cross-

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dataset benchmarking and alternative imputation methods (e.g., KNN, MICE) will be explored to enhance generalizability and model robustness.

5. Conclusions

Capturing complex behavioral patterns and enhancing marketing approaches were the issues presented in predicting repurchase behavior and optimizing marketing for e-commerce users. This research demonstrates that carefully integrating established DL models with adaptive evolutionary optimization can yield substantial performance gains in e-commerce repurchase prediction and marketing optimization. The predicting e-commerce user repurchase behavior data was obtained from the open source Kaggle, and the obtained information was preprocessed by handling missing values with imputation, duplicate removal, and min-max normalization to determine the numerical values presented in the research. An integrated DGA-EVMN was developed that incorporated the Vanilla from the Vanilla CNN approach with ELSTM (EVMN) for repurchasing behavior forecasting, and the DGA approach fine-tuned the hyperparameters of the EVMN approach. Experimental results demonstrated enhanced accuracy (0.984), precision (0.958), F1-score (0.960), AUC-ROC (0.971), and recall (0.962). The findings demonstrated the possibility of integrating predictive modeling with evolutionary optimization to provide intelligent, data-driven marketing techniques that assisted the enterprises in optimizing the value of a customer through adaptable and cost-effective techniques.

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Real-World Applicability

The proposed DGA-EVMN framework demonstrates high predictive accuracy and optimized marketing performance; however, practical deployment requires consideration of real-world constraints. Model inference latency is influenced by the combined Vanilla CNN and ELSTM layers, which may necessitate efficient hardware or model compression for real-time applications. Interpretability of the predictions is facilitated

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through the hierarchical feature extraction of Vanilla CNN and temporal modeling of ELSTM, but further explainable AI techniques could enhance marketing decision transparency. Despite these considerations, the framework is suitable for deployment in e-commerce platforms for adaptive marketing strategies, customer retention analysis, and personalized recommendation systems, provided sufficient computational resources and high-quality user data are available. Future work may focus on lightweight model variants, online learning capabilities, and integration with edge computing for low-latency applications.

Real-World Scenarios

The proposed DGA-EVMN framework can be applied in practical e-commerce and retail settings where understanding customer repurchase behavior is critical. Examples include:

- **Online marketplaces** (Amazon, eBay) for targeted marketing and personalized promotions.
- **Subscription-based services** (streaming, meal kits) to predict renewals and reduce churn.
- **Retail loyalty programs** to optimize rewards and retention strategies.
- **Digital marketing campaigns** to allocate discounts or offers cost-effectively based on predicted repurchase likelihood.

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Conflicts of interest

The author declares that this article is free of conflict of interest.

Data availability statement

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The data is in the article.

Ethics statement

Not applicable.

Clinical trial number

Not applicable.

Author contributions

Yang. Data curation&Writing—original draft preparation&Project administration

Consent to participate declarations

Not applicable.

Consent to publish declarations

Not applicable.

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